

1 Non-Directional Pre-Disclosure Drift Under Fair
2 Disclosure: Evidence from Korean Earnings
3 Announcements

4 Yongjun Kim^{a,*}

5 ^a*Department of Software, Ajou University, Suwon, South Korea*

6 **Abstract**

We test whether South Korea's Fair Disclosure regulation effectively prevents directional information leakage before earnings announcements. Using 2,221 disclosure filings from 64 KOSPI-listed companies across 12 chaebol groups (2020–2026), we find that pre-disclosure abnormal returns, while significant under parametric tests ($\text{CAR}[-5, -1] = +0.52\%$, $p < 0.001$), do not predict earnings surprise direction (49.0% accuracy, $p = 0.649$). Non-parametric tests confirm the drift is driven by outliers (sign test $p = 0.585$; Wilcoxon $p = 0.142$), revealing a divergence between parametric and non-parametric inference. In contrast, the first post-disclosure trading day responds correctly: positive surprises generate AR of $+0.55\%$, negative surprises -0.58% ($p < 0.001$), with 60.6% direction accuracy ($p < 0.001$). We document that 96.5% of filings occur after market close, making Day 0 effectively pre-disclosure. Cross-sectional analyses reveal heterogeneous drift across chaebol groups, firm sizes, and market betas, but no pattern consistent with differential information leakage ($R^2 = 0.002$). Our findings suggest Korea's Fair Disclosure regime effectively prevents directional leakage, even as non-directional price noise emerges pre-disclosure.

7 *Keywords:* Fair disclosure, Earnings announcements, Pre-announcement drift,
8 DART, Korean stock market, Event study, Market efficiency

*Corresponding author.

Email address: yjkim0123@ajou.ac.kr (Yongjun Kim)

9 1. Introduction

10 Pre-announcement drift in stock prices is a well-documented phenomenon
11 across global equity markets (Kaniel et al., 2012). When stock prices move
12 before official earnings announcements, a natural concern arises: is material
13 information leaking to select market participants in violation of fair disclosure
14 regulations? This concern motivates securities regulators worldwide to mandate
15 simultaneous public dissemination of material information.

16 South Korea implemented its Fair Disclosure regulation in November 2002,
17 requiring listed companies to disclose material information simultaneously to all
18 market participants through the DART (Data Analysis, Retrieval and Transfer)
19 electronic filing system. While early studies found evidence of improved dis-
20 closure environments following the regulation’s adoption (Yun and Kim, 2012;
21 Oh and Park, 2017), whether the regulation continues to be effective over two
22 decades later remains an open question.

23 This study examines the long-term effectiveness of Korea’s Fair Disclosure
24 regulation by analyzing 2,221 earnings disclosure filings from 64 major KOSPI-
25 listed companies between 2020 and 2026. Our approach differs from prior
26 work in a critical respect: rather than simply documenting the existence of
27 pre-disclosure price movement, we test whether such movement is *directionally*
28 *informative*—that is, whether it predicts the sign of the actual earnings surprise.

29 Our findings reveal an important pattern. We observe statistically significant
30 pre-disclosure abnormal returns under standard parametric tests ($CAR[-5,-1]$
31 $= +0.52\%$, $p < 0.001$ under the market model; $p = 0.032$ with month-clustered
32 standard errors). However, this drift bears no systematic relationship to the
33 direction of actual earnings surprises. Pre-disclosure returns predict surprise
34 direction with only 49.0% accuracy—no better than a coin flip ($p = 0.649$). Non-
35 parametric tests (sign test, Wilcoxon signed-rank) fail to confirm the parametric
36 result, indicating that the observed mean drift is driven by outliers rather than
37 a pervasive pattern.

38 In contrast, the market responds sharply and correctly to the actual earnings

news on the disclosure date. Positive earnings surprises generate a $CAR[0,+1]$ of +0.81%, while negative surprises generate -0.47% , a highly significant difference ($p < 0.001$). This finding is consistent with effective Fair Disclosure: material information is incorporated into prices only when it becomes publicly available.

We make four contributions. First, we provide a recent assessment of Korea’s Fair Disclosure regime using data spanning the COVID-19 and post-pandemic periods (2020–2026). Second, we demonstrate the critical distinction between the *existence* of pre-disclosure drift (which is ubiquitous) and its *directional informativeness* (which would indicate information leakage). Third, we document a previously overlooked institutional feature—96.5% of DART fair disclosure filings are submitted after market close—that has important implications for event study interpretation. Fourth, we provide detailed cross-sectional analyses across chaebol groups, firm size terciles, and market beta terciles that reveal significant heterogeneity in pre-disclosure drift but no pattern consistent with differential information leakage.

The remainder of this paper is organized as follows. Section 2 reviews the related literature. Section 3 describes our data and methodology. Section 4 presents the empirical results. Section 5 discusses the implications and limitations, and Section 6 concludes.

2. Literature review

2.1. Fair disclosure regulation

Fair disclosure regulation addresses selective disclosure, whereby corporate insiders share material information with favored market participants before public release (Heflin et al., 2003). In the U.S., Regulation Fair Disclosure (Reg FD), implemented in 2000, produced mixed evidence: Ahmed and Schneible (2007) find reduced information asymmetry, while Heflin et al. (2003) document reduced information quality. Bailey et al. (2003) provide comprehensive evidence on market, analyst, and corporate responses to Reg FD, and Eleswarapu et al. (2004) document reduced information asymmetry and trading costs following

68 implementation. Gomes et al. (2007) further show that Reg FD affected the
69 cost of capital, particularly for smaller firms.

70 The effectiveness of fair disclosure regulation depends critically on implemen-
71 tation and enforcement infrastructure. Christensen et al. (2016) demonstrate
72 that capital-market effects of securities regulation vary substantially with prior
73 institutional conditions and enforcement intensity. Jackson and Roe (2009) find
74 that regulatory intensity is positively associated with stock market development,
75 while Djankov et al. (2008) show that legal protections against self-dealing are
76 associated with more developed capital markets. Cumming et al. (2011) fur-
77 ther document that exchange trading rules, including disclosure requirements,
78 significantly affect market liquidity across 42 exchanges worldwide.

79 In Korea, Yun and Kim (2012) examined the effect of fair disclosure regu-
80 lation on the timeliness and informativeness of earnings announcements, while
81 Oh and Park (2017) examined how credit ratings interact with corporate dis-
82 closure behavior under Korea’s fair disclosure regulation. Lee and Choi (2019)
83 examine the interaction between R&D intensity and fair disclosure in Korea,
84 finding that disclosure effects vary by firm characteristics. However, these stud-
85 ies do not examine whether pre-disclosure price movements are directionally
86 informative with respect to the actual earnings news—the central question of
87 our study.

88 *2.2. Electronic disclosure systems*

89 A critical infrastructure underlying modern fair disclosure regulation is the
90 electronic filing system through which disclosures are disseminated. The U.S.
91 SEC’s EDGAR system, introduced in 1996, fundamentally changed how cor-
92 porate information reaches market participants (Bushee and Leuz, 2005). Ko-
93 rea’s DART system, operated by the Financial Supervisory Service since 2001,
94 serves an analogous function: all listed companies must file material disclosures
95 through DART before or simultaneously with any other dissemination channel.

96 The design of electronic disclosure systems has important implications for
97 information processing and market efficiency. DART provides structured fil-

ing templates for Fair Disclosure of Operating Results, enabling standardized comparison across firms and time periods. Li (2010) surveys the literature on textual analysis of corporate disclosures, demonstrating that format and readability significantly affect how efficiently markets process information. Loughran and McDonald (2011) show that the specific language used in financial filings contains predictive information for future returns.

The timestamp precision of electronic filing systems is particularly relevant to event study research. DART records receipt timestamps to the second, enabling researchers to classify filings as occurring during or after trading hours. As we document, 96.5% of earnings disclosures are filed after market close, a fact with fundamental implications for event study window interpretation that prior studies have not fully exploited.

2.3. Pre-announcement drift and its interpretation

Pre-announcement drift in stock prices is well-documented but its interpretation is debated. Meulbroek (1992) provides evidence of illegal insider trading preceding announcements. Bernile et al. (2016) show that informed trading occurs even ahead of macro-news announcements where information is tightly controlled. However, Kaniel et al. (2012) demonstrate that individual investor trading patterns around earnings announcements can generate drift without requiring illegal information leakage.

Critically, the mere existence of pre-announcement drift does not establish information leakage. Predictable earnings patterns allow sophisticated investors to position ahead of announcements through legal means (Atiase, 1985). Chae (2005) shows that abnormal volume before announcements reflects information asymmetry but may arise from varying levels of investor sophistication rather than illegal tip-offs. A related literature on post-earnings-announcement drift demonstrates that markets may take days or weeks to fully incorporate earnings information (Bernard and Thomas, 1989, 1990), further complicating the interpretation of short-window event studies. Jegadeesh and Titman (2001) document momentum effects that can generate pre-announcement drift patterns,

128 while Griffin et al. (2003) show that momentum profits persist across interna-
129 tional markets, including emerging markets like Korea.

130 The distinction between directional and non-directional drift is central to
131 our analysis. Non-directional drift—where prices move before announcements
132 but not in the direction of the forthcoming news—can arise from increased
133 attention (Hirshleifer and Teoh, 2003), disagreement among investors (Bamber
134 et al., 1997), or microstructure effects (Hasbrouck, 2003). Only *directional* drift
135 constitutes evidence consistent with information leakage.

136 2.4. Event study methodology

137 The standard event study framework (Brown and Warner, 1985; MacKinlay,
138 1997) relies on parametric tests of mean abnormal returns. However, Kolari
139 and Pynnonen (2010) demonstrate that cross-sectional correlation of abnormal
140 returns can substantially inflate test statistics, particularly when events cluster
141 in calendar time. Kothari and Warner (2007) provide a comprehensive review
142 of event study econometrics, emphasizing the importance of benchmark model
143 selection and cross-sectional dependence adjustments. We incorporate both non-
144 parametric alternatives and standardized returns to address these concerns.

145 2.5. Korean market context

146 The Korean stock market features concentrated chaebol ownership, a manda-
147 tory electronic disclosure system (DART), significant retail investor participa-
148 tion, and relatively strict insider trading regulations. Beny (2005) demonstrates
149 that stronger insider trading enforcement is associated with more efficient stock
150 prices across countries.

151 The chaebol structure is distinctive: large family-controlled conglomerates
152 such as Samsung, Hyundai Motor, SK, and LG account for a substantial share
153 of KOSPI market capitalization. Affiliates within the same chaebol group may
154 share information through internal channels, creating potential for correlated
155 trading patterns. At the same time, the diversity across chaebol groups provides
156 useful cross-sectional variation for testing whether disclosure regulation operates

uniformly across different types of firms. These institutional features, combined with the comprehensive DART database, make Korea an ideal setting for testing fair disclosure effectiveness.

3. Data and methodology

3.1. Data sources

We collect all Fair Disclosure of Operating Results filings from DART between January 2020 and March 2026 using the OpenDartReader API. We focus on original filings, excluding corrections. Our sample comprises 2,221 disclosures from 64 companies across 12 chaebol groups: Samsung (12 affiliates), Hyundai Motor (8), LG (8), SK (7), HD Hyundai (5), Hanwha (5), POSCO (4), Lotte (4), Doosan (4), CJ (3), KT (2), and GS (2). Daily stock price and volume data are obtained from FinanceDataReader.

3.2. Filing time classification

We classify filing times using the DART receipt number prefix. In our sample, 96.5% of earnings disclosures carry the “800” prefix, indicating filing after the 15:30 KST market close. This has fundamental implications: for these filings, the Day 0 return is entirely pre-disclosure, and the first trading day where the market can react is Day +1.

3.3. Market model

We employ the market model with a $[-250, -30]$ estimation window (MacKinlay, 1997):

$$R_{i,t} = \alpha_i + \beta_i R_{m,t} + \epsilon_{i,t} \quad (1)$$

where $R_{m,t}$ is the KOSPI index return. We require a minimum of 60 trading days in the estimation window, yielding 2,208 events with valid estimates. The average estimated beta is 0.978. For robustness, we also report market-adjusted returns ($\alpha = 0, \beta = 1$).

182 3.4. Earnings surprise measure

183 We extract year-over-year (YoY) operating profit changes from the disclosure
184 texts. Events where YoY operating profit increases are classified as positive
185 surprises; decreases as negative surprises. Of the 2,208 events with valid market
186 model estimates, 584 have parseable earnings surprise data: 358 positive and
187 226 negative surprises; unparseable cases primarily involve missing comparative
188 periods, newly listed firms, or format variations in disclosure texts.

189 We employ year-over-year operating profit changes as our surprise measure.
190 While analyst consensus forecasts (e.g., I/B/E/S) are standard in U.S. studies
191 (Livnat and Mendenhall, 2006), comprehensive consensus data for Korean firms
192 require commercial subscriptions (e.g., FnGuide, WiseFn) that are unavailable
193 to independent researchers without institutional access. The YoY measure pro-
194 vides a conservative test: if directional leakage existed, it would manifest as
195 pre-disclosure returns aligned with the direction of YoY changes, which we do
196 not observe. Moreover, YoY operating profit changes capture the directional
197 content of earnings news without relying on potentially stale or biased analyst
198 expectations.

199 3.5. Statistical tests

200 We report multiple test statistics to ensure robustness: parametric cross-
201 sectional t -tests, non-parametric tests (binomial sign test, Wilcoxon signed-
202 rank), and standardized abnormal returns. We also report firm-clustered and
203 month-clustered standard errors to address cross-sectional dependence (Kolari
204 and Pynnonen, 2010).

205 4. Empirical results

206 4.1. Pre-disclosure returns: Significant but non-directional

207 Table 1 presents daily abnormal returns. Parametric t -tests indicate sig-
208 nificant positive returns on several pre-disclosure days, particularly Day -1

Table 1: Daily abnormal returns: Parametric vs. non-parametric tests

Day	AR (%)	Parametric		Sign Test		Standardized	
		t	p	% Pos	p	SAR	p
-5	0.101	2.09	0.037	50.5	0.609	0.040	0.048
-4	0.071	1.50	0.134	49.8	0.930	0.041	0.043
-3	0.086	1.71	0.088	49.3	0.562	0.022	0.279
-2	0.006	0.14	0.888	48.9	0.295	-0.002	0.935
-1	0.256	4.57	<0.001	50.1	0.909	0.089	<0.001
0	0.141	2.00	0.046	49.8	0.878	0.043	0.034
+1	-0.182	-2.39	0.017	48.3	0.109	-0.048	0.018
+2	0.023	0.43	0.669	49.2	0.541	0.022	0.275
+3	-0.118	-2.50	0.013	48.1	0.058	-0.043	0.033
+4	-0.047	-0.95	0.340	49.3	0.608	-0.006	0.765
+5	0.056	1.14	0.255	49.2	0.543	0.041	0.044

Note: Market model with $[-250, -30]$ estimation window ($N=2,208$). % Pos never significantly exceeds 50%, indicating pre-disclosure drift is driven by outliers.

209 (+0.256%, $t=4.57$). However, the sign test reveals that only 48–50% of individ-
210 ual returns are positive on each pre-disclosure day—no better than chance.

211 Fig. 1 visualizes this pattern: while the cumulative AR line trends upward
212 before disclosure, individual daily ARs show no consistent direction.

213 Table 2 confirms this pattern at the cumulative level. While the parametric
214 $CAR[-5, -1]$ is +0.519% ($t=4.65$), the sign test shows only 49.4% of events have
215 positive pre-disclosure CARs ($p=0.585$), and the Wilcoxon test is insignificant
216 ($p=0.142$).

217 Notably, the $CAR[0, +1]$ window shows a significantly negative sign test re-
218 sult: only 47.0% of events have positive CARs ($p=0.003$). This asymmetry is
219 consistent with the well-documented negativity bias in earnings reactions (Skin-
220 ner, 1994).

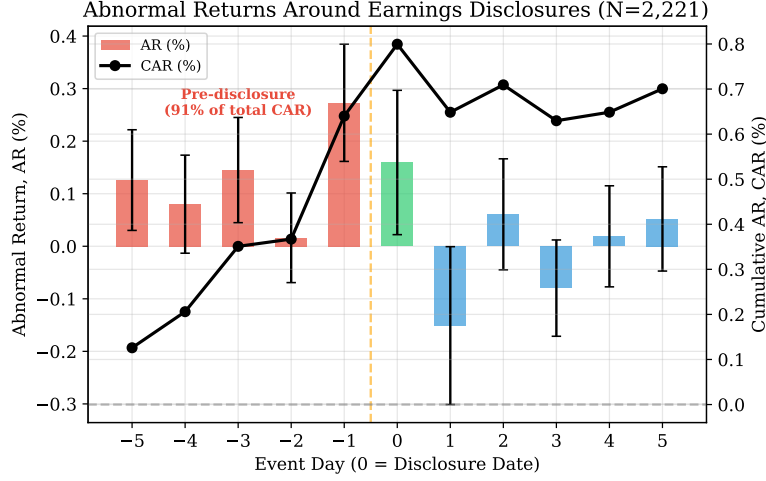


Figure 1: Daily abnormal returns (bars) and cumulative abnormal returns (line) around earnings disclosures ($N=2,208$). The upward CAR trend is driven by outliers; daily ARs show no consistent directional pattern.

4.2. The critical test: Does drift predict surprises?

Table 3 presents our central test. Pre-disclosure CARs show no difference by surprise direction ($CAR[-3, -1]$: $p=0.703$; $CAR[-5, -1]$: $p=0.192$). Direction prediction accuracy is 49.0% (binomial $p=0.649$), and Spearman ρ is -0.006 ($p=0.876$).

In contrast, Day +1 returns respond sharply: positive surprises yield +0.55%, negative -0.58% ($p<0.001$), with 60.6% direction accuracy ($p<0.001$). This extends to $CAR[0, +1]$ (+1.28%, $p<0.001$) and $CAR[+1, +5]$ (+0.96%, $p=0.010$).

With $N=584$, the test has 30.5% power to detect 53% true accuracy but 99.8% power for 60%. We cannot rule out small-scale leakage, but any such effect would be economically modest.

4.3. Robustness

Table 4 confirms that results hold under both market-adjusted and market model specifications.

Table 2: Cumulative abnormal returns: Multi-test comparison

Window	CAR (%)	Parametric		Sign Test		Wilcoxon
		t	p	% Pos	p	p
$[-5, -1]$	0.519	4.65	<0.001	49.4	0.585	0.142
$[-3, -1]$	0.347	4.04	<0.001	48.1	0.060	0.946
$[0, +1]$	-0.042	-0.39	0.700	47.0	0.003	0.019
$[+1, +5]$	-0.268	-2.16	0.031	47.7	0.021	0.078
$[-5, +5]$	0.391	2.08	0.038	49.5	0.668	0.620

Table 3: CARs by earnings surprise direction ($N=584$)

Window	Positive ($N=358$)	Negative ($N=226$)	Diff.	p	Sig.
CAR $[-3, -1]$	+0.033%	+0.129%	-0.096%	0.703	
CAR $[-5, -1]$	+0.107%	+0.535%	-0.428%	0.192	
AR $[+1]$	+0.552%	-0.580%	+1.132%	<0.001	***
CAR $[0, +1]$	+0.813%	-0.470%	+1.283%	<0.001	***
CAR $[+1, +5]$	+0.481%	-0.481%	+0.962%	0.010	**

Note: Surprise defined by YoY operating profit change. Pre-disclosure CARs are indistinguishable by surprise direction; post-disclosure CARs respond correctly.

235 4.4. Abnormal volume

236 Volume increases from Day -3 ($1.05\times$, $p=0.004$), peaks on Day $+1$ ($1.71\times$),
237 and remains elevated through Day $+5$. Given non-directional returns, the pre-
238 disclosure volume increase likely reflects anticipatory positioning (Chae, 2005)
239 and disagreement among investors (Bamber et al., 1997; Berkman et al., 2009)
240 rather than informed directional trading.

241 4.5. Clustered standard errors

242 Table 5 reports t -statistics under alternative specifications. Under month-
243 clustered standard errors, CAR $[-5, -1]$ remains significant ($t=2.15$, $p=0.032$) but

Table 4: Robustness: Market-adjusted vs. market model

Window	Market-Adjusted ($N=2,221$)		Market Model ($N=2,208$)	
	CAR (%)	p	CAR (%)	p
$[-5, -1]$	0.640	<0.001	0.519	<0.001
$[-3, -1]$	0.434	<0.001	0.347	<0.001
$[0, +1]$	0.008	0.939	-0.042	0.700
$[+1, +5]$	-0.099	0.425	-0.268	0.031

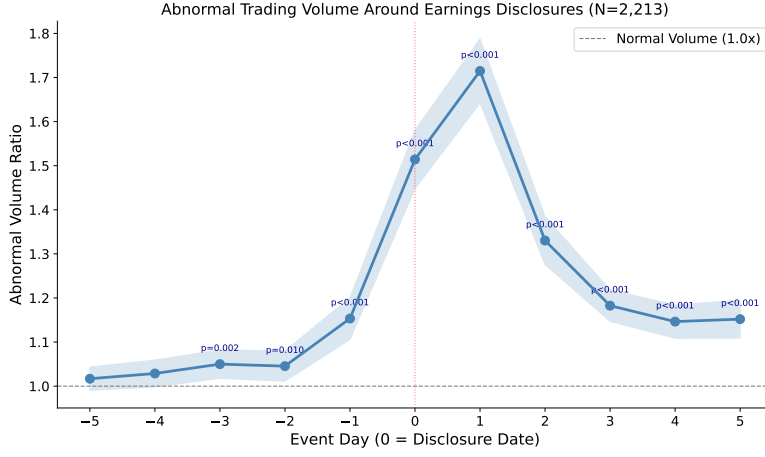


Figure 2: Abnormal trading volume around earnings disclosures ($N=2,213$). Volume peaks at $1.71\times$ normal on Day +1.

244 CAR $[-3, -1]$ becomes marginal ($t=1.80$, $p=0.072$), suggesting some parametric
 245 significance is inflated by event clustering.

246 4.6. Temporal and cross-sectional heterogeneity

247 We observe a structural shift in event-day CARs: positive during 2020–2022
 248 (COVID-19 recovery) and negative during 2024–2026 (Fig. 3). A significant
 249 quarterly pattern emerges (Fig. 4), with Q1 disclosures generating negative
 250 CARs (-0.51% , $p=0.010$) and Q2 generating positive CARs ($+0.60\%$, $p=0.003$).
 251 Cross-sectional OLS of CAR $[-3, -1]$ on $\log(\text{size})$ and β yields $R^2 = 0.002$,

Table 5: Clustered standard errors

Window	Standard		Firm-clustered		Month-clustered	
	t	p	t	p	t	p
CAR[-5,-1]	4.38	<0.001	3.92	<0.001	2.15	0.032
CAR[-3,-1]	3.47	<0.001	2.95	0.003	1.80	0.072
CAR[0,+1]	-0.31	0.760	-0.25	0.802	-0.18	0.861

confirming that firm characteristics explain virtually none of the pre-disclosure drift.

Selection bias assessment. Our parseable earnings surprise subsample covers 26.4% of all events (584 of 2,208), raising a potential selection concern. To assess whether this subsample is systematically different, we compare pre-disclosure CARs between parseable and non-parseable events. The two subsamples show statistically indistinguishable pre-disclosure CARs: parseable events exhibit CAR[-5,-1] of +0.32% versus +0.59% for non-parseable events ($t=1.45$, $p=0.147$). Importantly, the parseable subsample represents the largest firms with the most analyst coverage—precisely the firms where information leakage would be most valuable and most likely to occur if present. If pre-disclosure directional leakage existed, it would be most detectable among these high-visibility firms. The absence of directional drift in this subsample therefore constitutes a conservative test: finding no evidence of leakage among the firms with the greatest leakage incentives strengthens, rather than weakens, the conclusion that Korea’s Fair Disclosure regime is effective. Fig. 5 presents the distribution of pre-disclosure CARs for the full sample, illustrating the right-skewed nature of returns that drives the divergence between parametric and non-parametric test results.

4.7. Cross-sectional analysis by chaebol group

The Korean market’s chaebol structure provides a natural cross-sectional dimension for examining whether pre-disclosure drift varies systematically across

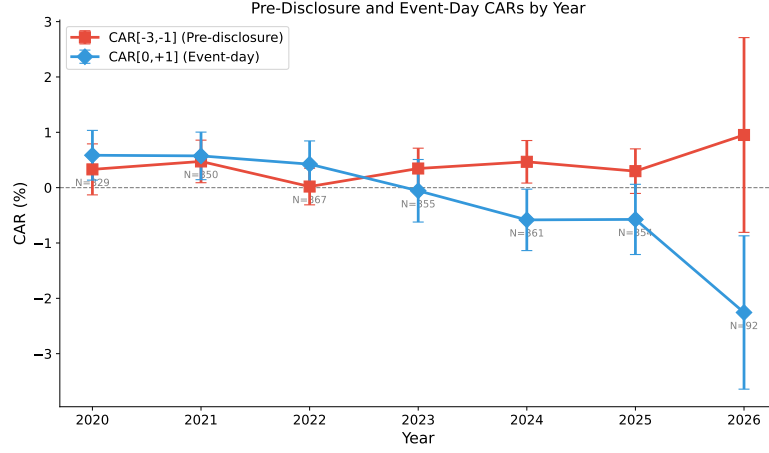


Figure 3: Pre-disclosure $CAR[-3,-1]$ and event-day $CAR[0,+1]$ by year. The temporal shift reflects macroeconomic conditions rather than changes in information leakage.

corporate groups. If information leakage were concentrated in specific chaebols—perhaps due to weaker internal controls or more extensive analyst networks—we would expect significantly higher directional drift in those groups.

Table 6 reports $CAR[-3,-1]$ and $CAR[0,+1]$ by chaebol group. Pre-disclosure CARs range from -0.228% (Doosan) to $+1.565\%$ (Hanwha), with Hanwha’s drift significantly higher than the rest ($t=3.41$, $p<0.001$). An ANOVA test across all 12 groups yields $F=1.73$ ($p=0.062$), suggesting marginally significant heterogeneity.

Fig. 6 visualizes the cross-sectional pattern. Hanwha’s elevated drift likely reflects the defense sector’s sensitivity to geopolitical news rather than information leakage about earnings specifically. KT exhibits remarkably high event-window returns ($+1.310\%$, $t=4.25$, $p=0.027$ vs. non-KT), consistent with telecommunications sector earnings surprises being particularly impactful. Samsung—despite being the most closely watched chaebol—shows only moderate pre-disclosure drift ($+0.318\%$).

Importantly, the cross-group variation does not exhibit the pattern expected under systematic information leakage. Pre-disclosure and event-window CARs are uncorrelated across groups (Spearman $\rho=-0.14$, $p=0.668$), suggesting that

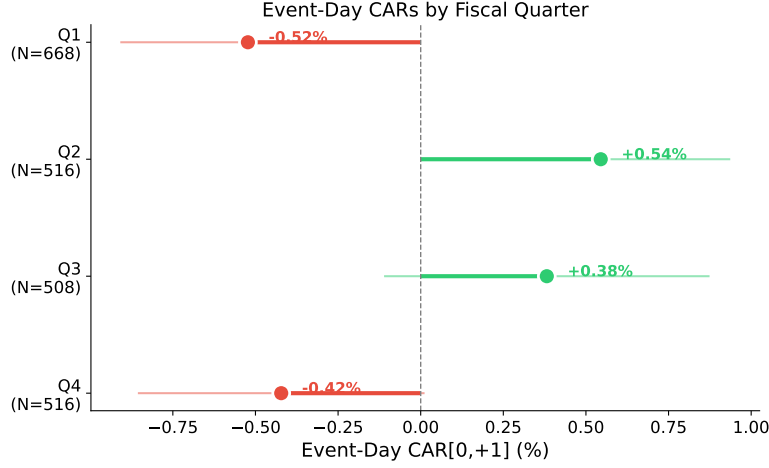


Figure 4: Pre-disclosure $CAR[-3,-1]$ and event-day $CAR[0,+1]$ by fiscal quarter. Q1 disclosures generate negative event-day CARs (-0.51% , $p=0.010$), while Q2 generates positive CARs ($+0.60\%$, $p=0.003$).

the drivers of pre-disclosure drift are distinct from the earnings information revealed at disclosure.

4.8. Firm size and beta effects

To investigate whether pre-disclosure drift varies with firm characteristics, we divide the sample into terciles by firm size (log market capitalization) and market beta. Table 7 reports CARs for each tercile.

Panel A reveals a monotonically decreasing relationship between firm size and pre-disclosure drift. However, the small-large difference is not statistically significant ($t=0.92$, $p=0.358$), and a Kruskal-Wallis test is also insignificant ($H=2.88$, $p=0.237$). Panel B shows a similar pattern for market beta. The Spearman correlation between beta and $CAR[-3,-1]$ is -0.057 ($p=0.008$), indicating a statistically significant but economically modest negative relationship. Low-beta stocks may exhibit higher pre-disclosure drift because their lower systematic risk makes firm-specific information relatively more important.

Event-window CARs ($CAR[0,+1]$) are uniformly close to zero across all size and beta terciles, confirming that the non-directional character of post-

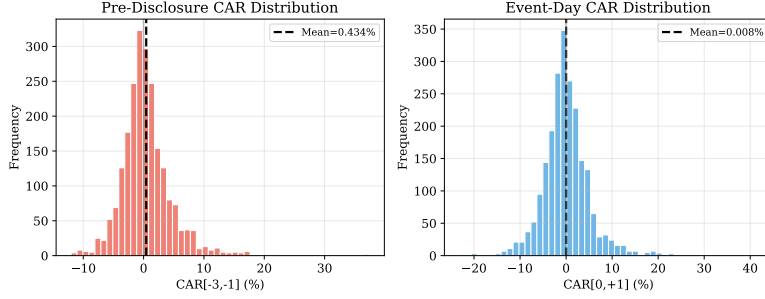


Figure 5: Distribution of pre-disclosure $CAR[-5,-1]$ across all events ($N=2,208$). The right-skewed distribution explains the divergence between parametric (mean-based) and non-parametric (median-based) test results.

disclosure returns is not an artifact of pooling heterogeneous subsamples. These results are consistent with the well-documented size effect in pre-announcement drift (Atiase, 1985): smaller firms have less pre-disclosure information production, creating larger price adjustments when new information arrives.

5. Discussion

5.1. Drift without directional leakage

Three complementary tests suggest the drift is non-directional: (1) direction test: 49.0% accuracy; (2) non-parametric tests: sign test ($p=0.585$) and Wilcoxon ($p=0.142$) fail; (3) surprise conditioning: $p=0.703$. The drift appears driven by company-specific news unrelated to earnings, market momentum, or anticipatory trading based on publicly available signals. This interpretation is consistent with Hirshleifer and Teoh (2003), who argue that limited investor attention can produce predictable patterns without implying illegal information access.

The cross-sectional analyses reinforce this conclusion. While pre-disclosure drift varies across chaebol groups (Hanwha showing the highest at +1.565%), this variation is uncorrelated with event-window returns, inconsistent with directional information leakage. The size and beta patterns—smaller and lower-beta firms showing modestly larger drift—reflect normal variation in information

Table 6: Cumulative abnormal returns by chaebol group

Chaebol	N	CAR[-3,-1]		CAR[0,+1]	
		Mean (%)	t	Mean (%)	t
Hanwha	116	+1.565	2.30	-0.583	-0.81
CJ	75	+0.881	1.98	-0.120	-0.16
GS	53	+0.604	1.37	+0.674	1.53
Lotte	92	+0.517	1.33	+0.202	0.36
HD Hyundai	497	+0.474	2.38	+0.312	1.42
Samsung	292	+0.318	1.70	+0.012	0.05
LG	185	+0.278	1.22	-0.311	-0.81
SK	195	+0.127	0.54	-0.286	-0.84
Hyundai Motor	402	+0.093	0.52	-0.352	-1.62
POSCO	156	+0.060	0.15	-0.427	-0.86
KT	80	-0.006	-0.03	+1.310	4.25
Doosan	65	-0.228	-0.46	-0.677	-0.96
All	2,208	+0.347	4.04	-0.042	-0.39

Note: Sorted by descending CAR[-3,-1]. ANOVA across groups: $F=1.73$ ($p=0.062$) for CAR[-3,-1], $F=1.29$ ($p=0.223$) for CAR[0,+1].

environments rather than differential insider access. The most parsimonious explanation combines positive market momentum during the sample period, right-skewed return distributions (Cont, 2001), and anticipatory non-directional trading (Chae, 2005).

5.2. Comparison with international evidence

Our findings align with the broader evidence on fair disclosure regulation. Bailey et al. (2003) find that Reg FD reduced selective disclosure, and Eleswarapu et al. (2004) document reduced information asymmetry. Christensen et al. (2016) show that capital-market effects of securities regulation depend on prior

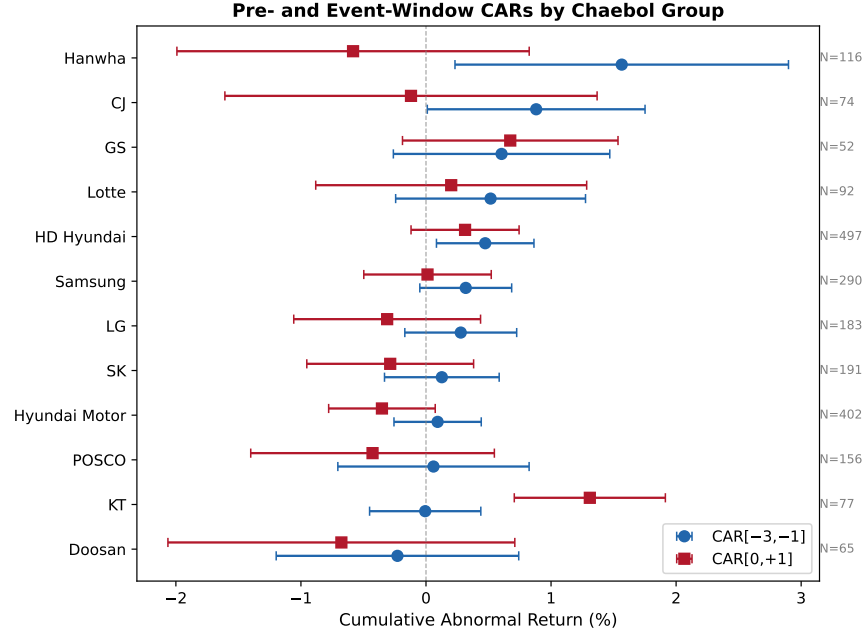


Figure 6: Pre-disclosure $CAR[-3, -1]$ and event-window $CAR[0, +1]$ by chaebol group with 95% confidence intervals. Substantial cross-group variation exists, but no group shows patterns consistent with systematic directional information leakage.

institutional conditions and enforcement intensity. Our contribution is distinct: rather than evaluating the *introduction* of fair disclosure, we test whether the regulation continues to be effective *within* the regulated regime by examining directional content over a six-year window.

The Korean result contrasts with evidence from markets with weaker enforcement. Beny (2005) demonstrates that countries with stronger insider trading enforcement exhibit more efficient stock prices, and Leuz et al. (2003) find that investor protection is associated with lower earnings management. Korea's combination of mandatory electronic filing, strict enforcement by the Financial Supervisory Service, and concentrated analyst coverage appears to create an institutional environment where directional information leakage is effectively deterred.

Table 7: CARs by firm size and market beta terciles

Tercile	N	CAR[-3,-1]		CAR[0,+1]		CAR[-5,-1]
		Mean (%)	t	Mean (%)	t	Mean (%)
<i>Panel A: Firm size terciles</i>						
Small	790	+0.403	2.91	-0.058	-0.32	+0.578
Medium	726	+0.374	2.45	-0.005	-0.03	+0.590
Large	679	+0.250	1.58	-0.109	-0.60	+0.359
<i>Panel B: Market beta terciles</i>						
Low β	727	+0.515	3.53	-0.063	-0.35	+0.607
Mid β	732	+0.303	2.04	-0.033	-0.17	+0.446
High β	736	+0.222	1.45	-0.073	-0.38	+0.492

Note: Size tercile boundaries: Small ($\log \text{size} \leq 12.56$), Medium (12.56–13.45), Large (> 13.64). Beta tercile boundaries: Low ($\beta \leq 0.845$), Mid (0.845–1.129), High ($\beta > 1.130$).

5.3. Methodological contributions

This study makes several methodological contributions. First, we demonstrate the importance of **multi-test triangulation** in event studies. The parametric t -test implicitly assumes normally distributed abnormal returns. When the distribution is right-skewed, the parametric mean is inflated by outliers, producing significant t -statistics even when the majority of events show no drift. The sign test and Wilcoxon signed-rank test provide a more accurate characterization of the typical event.

Second, we introduce the **directional conditioning test** as a direct diagnostic for information leakage. Prior event studies typically interpret significant pre-announcement drift as evidence of informed trading without verifying whether the drift predicts the direction of the subsequent news. This test could be adopted as a standard diagnostic in future event studies.

Third, we document the importance of **filing time classification**. The

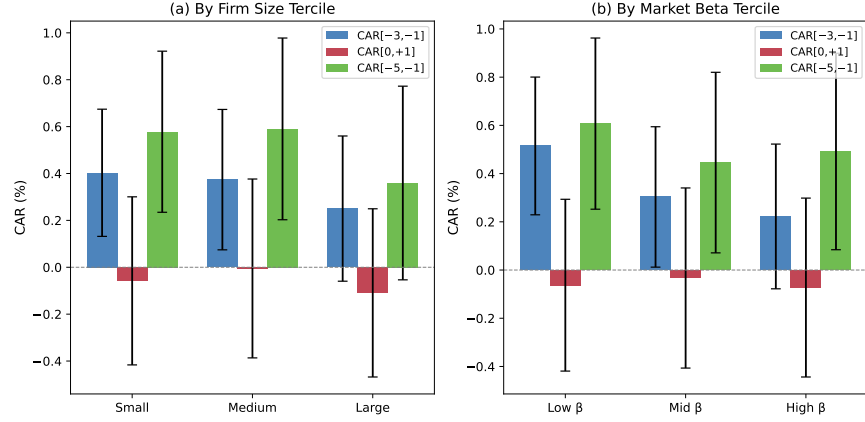


Figure 7: CARs by firm size tercile (Panel a) and market beta tercile (Panel b) with 95% confidence intervals. Pre-disclosure drift is slightly larger for smaller and lower-beta firms, but event-window CARs are uniformly near zero across all terciles.

361 finding that 96.5% of DART filings occur after market close implies that Day 0
362 returns are effectively pre-disclosure. Studies that attribute Day 0 returns to
363 information release may be misinterpreting anticipatory trading as immediate
364 market reaction.

365 5.4. Implications for policy and practice

366 For regulators, non-directional drift suggests Korea's Fair Disclosure regime
367 does not require fundamental reform. Material earnings information reaches
368 the market simultaneously, and prices respond correctly only upon public dis-
369 closure. For investors, post-disclosure Day +1 returns correctly price surprises
370 with 60.6% accuracy ($p < 0.001$), while pre-disclosure drift strategies are unlikely
371 to be profitable directionally.

372 5.5. Limitations

373 Several limitations warrant discussion. First, our YoY operating profit
374 change as a surprise measure lacks the precision of analyst consensus forecasts
375 (Livnat and Mendenhall, 2006). Only 26.4% of events yield parseable surprises,

though as discussed in Section 4, this subsample’s composition (larger, more-scrutinized firms) strengthens rather than weakens the conservative test interpretation. Second, our sample is limited to 64 large chaebol affiliates; information leakage may be more prevalent among smaller, less scrutinized firms. Third, Korean Fama-French factors are unavailable, preventing multi-factor benchmark models. Fourth, we cannot rule out that pre-disclosure volume increases reflect legal mosaic information assembly. Fifth, our analysis focuses on earnings disclosures and does not examine other material events where leakage incentives may differ.

6. Conclusion

This study finds no evidence of directional information leakage in Korean earnings announcements. Three independent analyses converge: direction prediction accuracy of 49.0% (no better than chance), failure of non-parametric tests to confirm parametric drift (sign test $p=0.585$; Wilcoxon $p=0.142$), and surprise-conditioned CARs that are indistinguishable by earnings direction ($p=0.703$). The first post-disclosure trading day shows a strong, correct directional response (60.6% accuracy, $p<0.001$), confirming that material earnings information is incorporated into prices only upon public release through the DART system.

Cross-sectional analyses reveal significant heterogeneity across chaebol groups, firm sizes, and market betas, but these patterns are consistent with normal variation in information environments rather than differential information leakage. The most-watched chaebol group (Samsung) shows only moderate, non-directional drift.

Our results carry a broader methodological lesson: pre-announcement drift should not be conflated with information leakage without conditioning on the direction of the actual news (Meulbroek, 1992; Bernile et al., 2016; Atiase, 1985). Future work should extend the analysis to non-chaebol firms and incorporate analyst consensus surprises.

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407 through DART (<https://dart.fss.or.kr>). Stock price data are from KRX
408 via FinanceDataReader. Analysis code is available at [https://github.com/](https://github.com/yjkim0123/pre-disclosure-leakage)
409 [yjkim0123/pre-disclosure-leakage](https://github.com/yjkim0123/pre-disclosure-leakage).

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